Digitalization: Standardization - Reliabile - Innovation

Digitalization: Critical Enablers

Key Parameters

- Standardization
- Reliable
- Innovation

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Digitalization Roadmap for Utilities

STANDARDIZATION MEASURES

- Smart Grid & Modern Grid
- Need for Interoperability
- Digitalization

RELIABILITY MEASURES

- Scalability
- Redundancy
- Protection

INNOVATION

- AI ML Integration
- Distributed Energy Resources , EV, BESS

Digitalization: **Standardization**, Reliability, Innovation

Digital-Ready Infrastructure: Connect Every Asset

Without standardization, digitalization becomes chaotic—multiple incompatible technologies create silos, inefficiencies, and procurement challenges.

Digitalization: **Standardization**, Reliability, Innovation

IT–OT Integration: The Digital Nerve Center for DISCOMs

PLCnext Technology

An Open, Scalable Automation Ecosystem

IEC61850/DNP3/ MPLS/MQTT

Ready Communication Platforms

Smart RTUs and Substation Automation Solutions

✓ This enables utilities to move from siloed operations toward real-time decision-making, loss reduction, and reliable supply management

DISCOM digitalization studies underline that IT–OT convergence is essential to unlock efficiency and visibility across metering, billing, SCADA, GIS, and outage systems

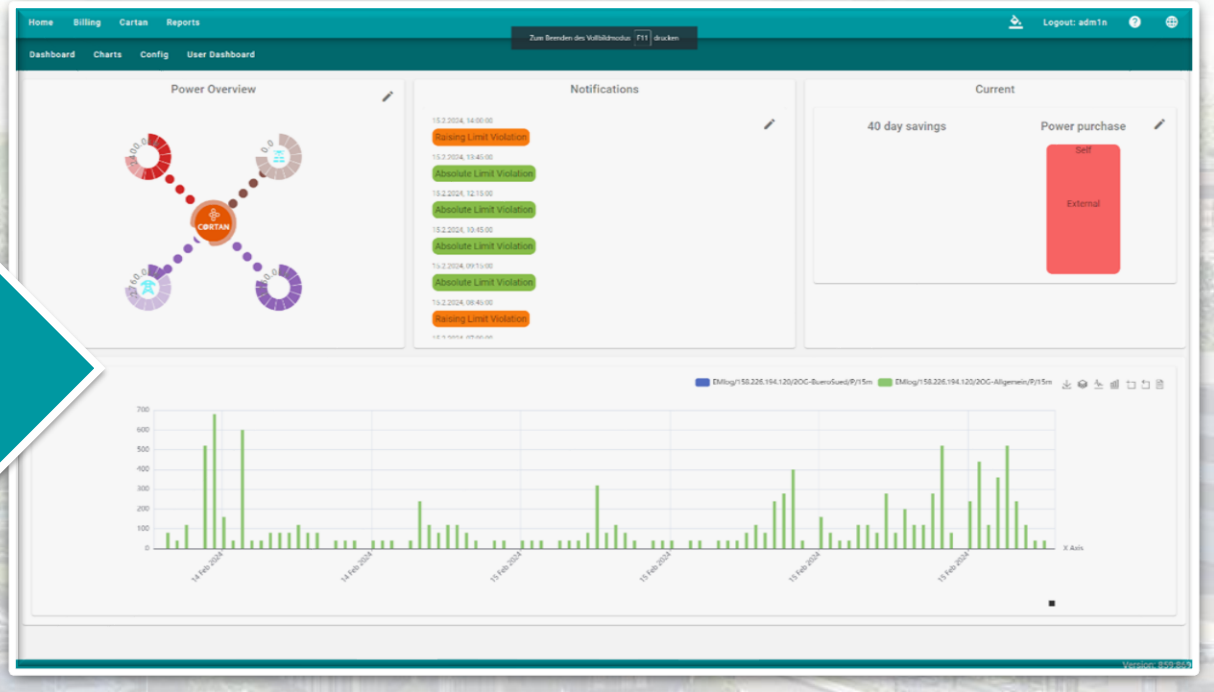
Digitalization: **Standardization**, Reliability, Innovation

Grid Automation & Smart Grid Modernization

Digital Substation

Feeder Automation

RMU Monitoring



These solutions directly support the long-term goals of RDSS and National Grid Modernization Agendas

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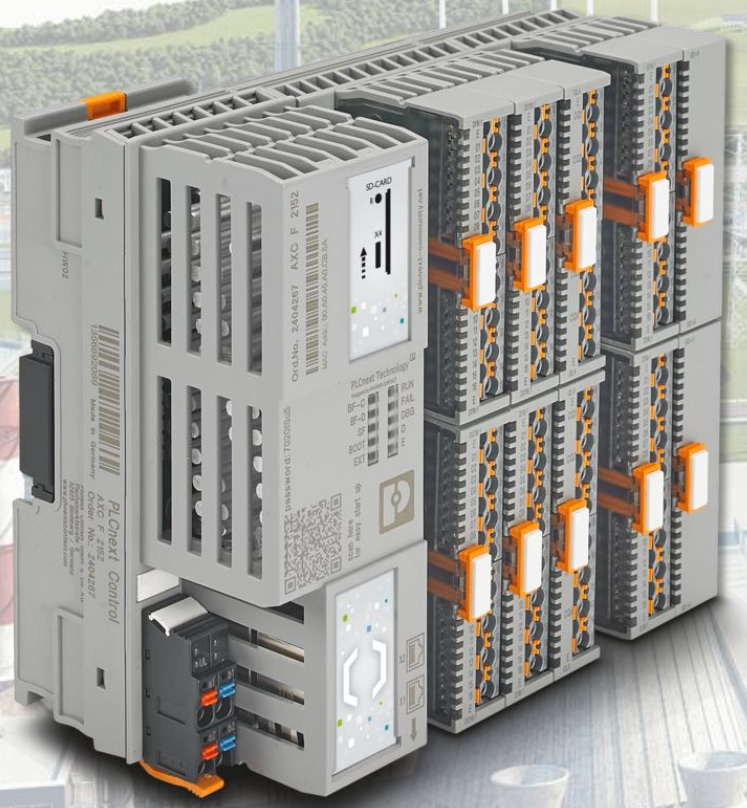
An Open System for Limitless Automation

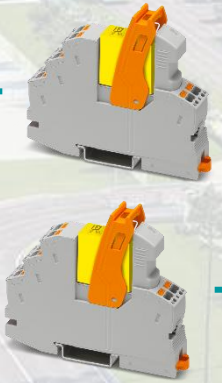


- ✓ **Implementation and acceptability of developer friendly language such as C, C++, Python, Json**
- ✓ **Openness of IEC protocols such as OPC UA, Profinet, Modbus, Profibus, DNP, IEC 60870-5-104 etc.**
- ✓ **Support Cloud/IT Functions like MQTT, SQL, SNMP, Rest API for Digitalization**
- ✓ **Scalable and Future oriented**

Redundant Protection Architecture

Ring Network

Redundant Relay

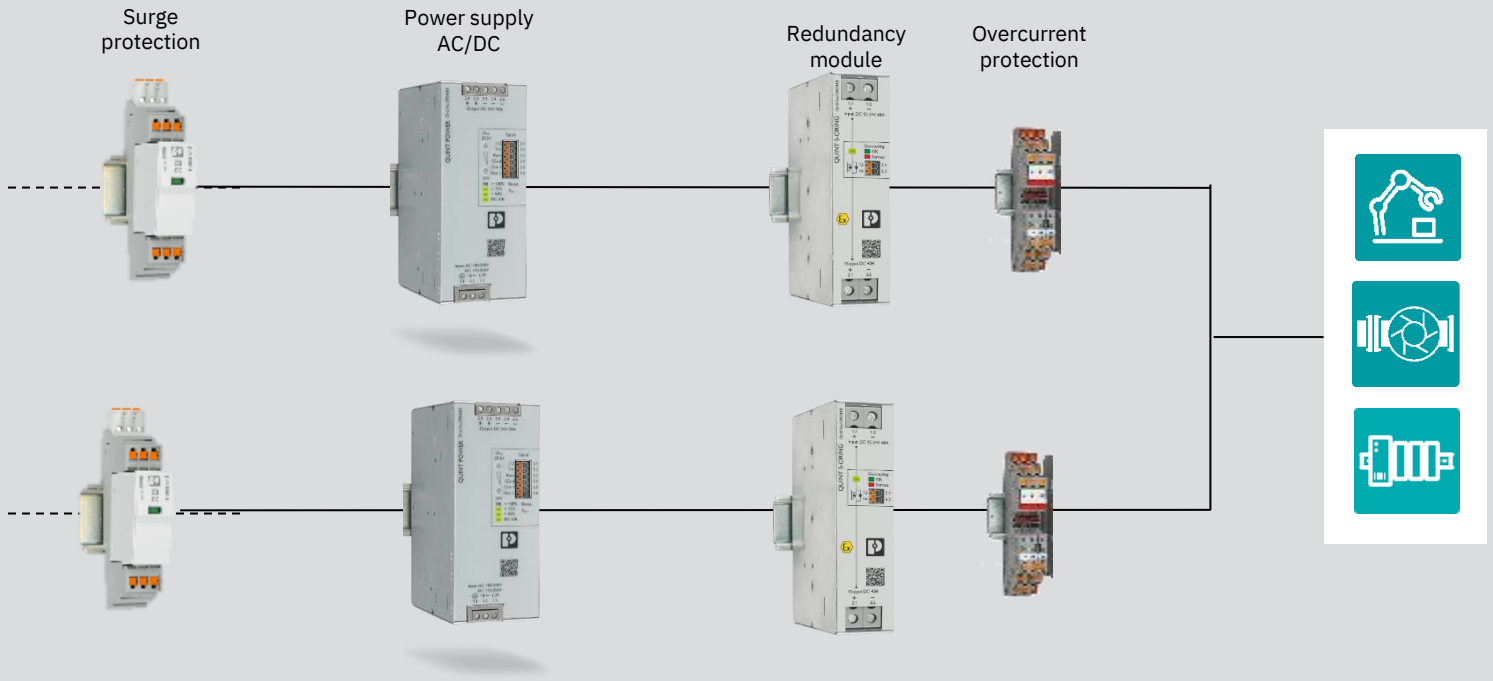


Breaker

- ✓ Redundant Protection for Sensitive Equipment
- ✓ Eliminate Failure of relay
- ✓ Ensure Safe Operations

Digitalization: Standardization, **Reliability,** Innovation

Power Redundancy



QUINT POWER power supply

QUINT S-ORING redundancy module

Surge protection PLUGTRAB-SEC

Electronic circuit breaker PTCB

Digitalization: Standardization, **Reliability,** Innovation

Lightning Protection

Surge protection for power supply
From the feed-in to the end device, our protective devices for power supply of type 1, type 2 and type 3 protect you against defects from lightning currents and overvoltage.

Type 1/2

Type 1+2

Type 2

Type 3

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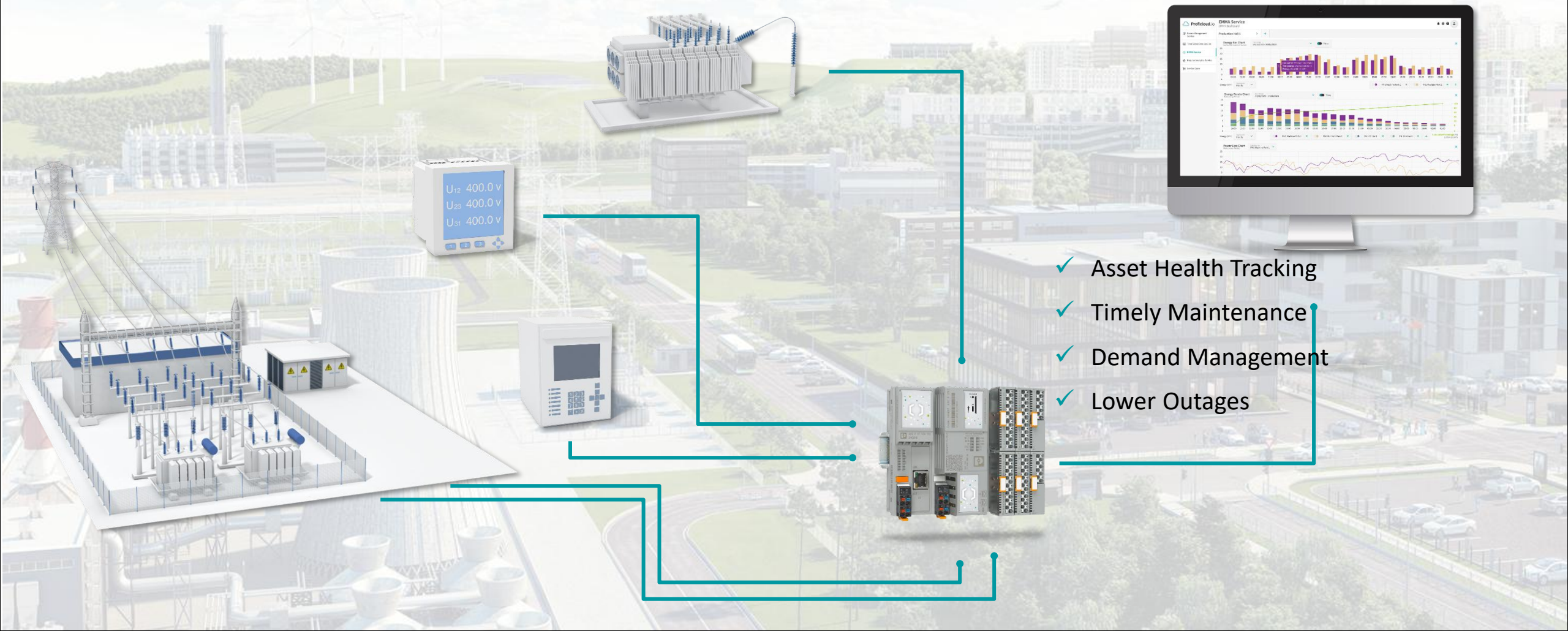
Digitalization: Standardization, Reliability, **Innovation**

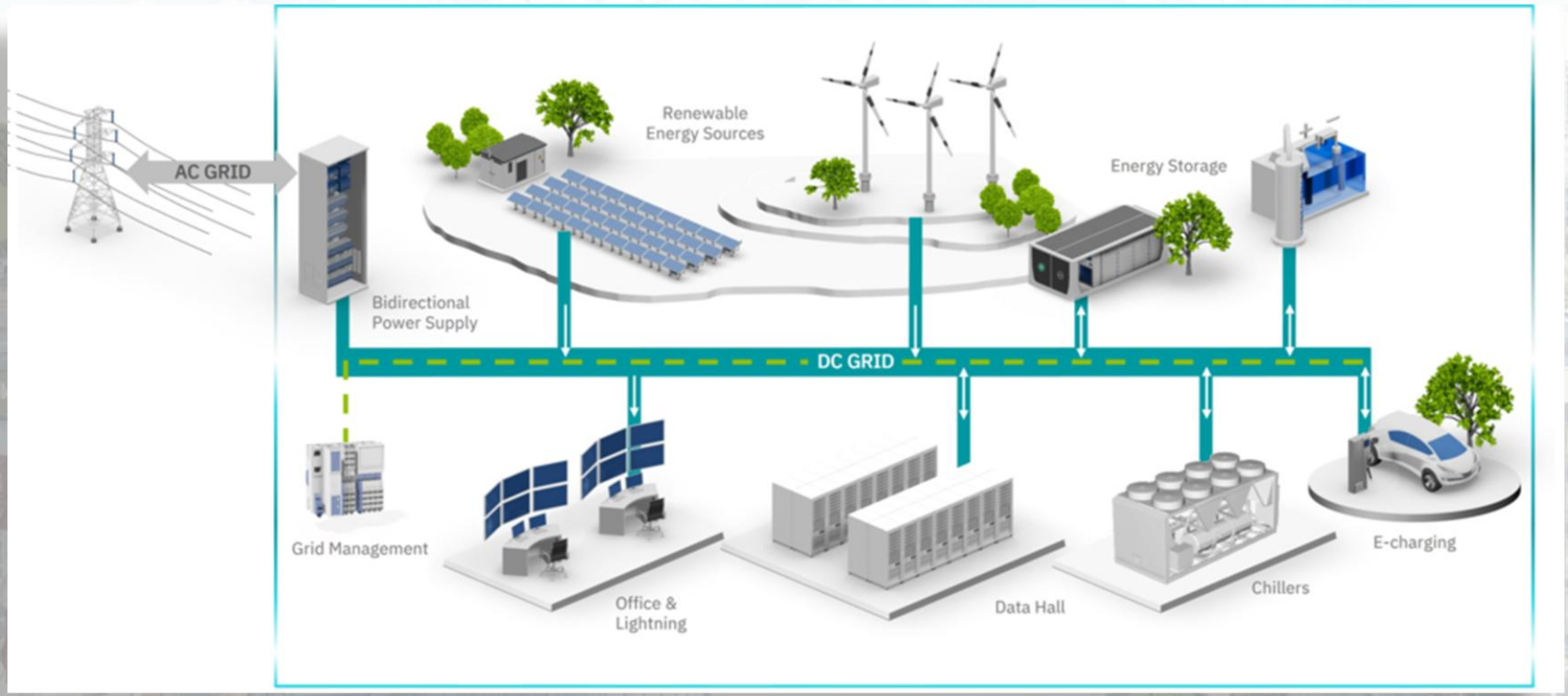
AI ML Integration for Predictive Maintenance



- ✓ Asset Health Tracking
- ✓ Timely Maintenance
- ✓ Demand Management
- ✓ Lower Outages

Digitalization: Standardization, Reliability, **Innovation**

Future-Ready Utility Architecture: DER, EV, Storage & Net Zero



- ✓ DER management and microgrid integration tools
- ✓ EV-ready distribution automation solutions
- ✓ Battery energy storage control systems
- ✓ Cybersecure, modular architectures built for growth

100 years of passion for technology and innovation

“ Phoenix Contact is a privately owned company founded in 1923 with great depths of added value. It is independent and has the freedom to make its own decisions as a company.

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Production sites

Germany | China | Taiwan | India | Poland | Sweden | Switzerland | Turkey | Argentina | Greece | USA

100,000

Products

22,000

Employees worldwide

75%

Sales abroad

25%

Sales in Germany

10,200

Employees in Germany

1923 > TODAY

Founded in Germany

Present in more than 100 countries

Group Executive Board:

Frank Stührenberg (CEO)
Axel Wachholz (CFO)
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Dirk Görlitzer (COO, President BA ICE)
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Over
100,000
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Cyber Security Need of an Hour

Meet us in Exhibition Hall & lets collaborate on making Digitalization Reliable, Available and Secured.

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